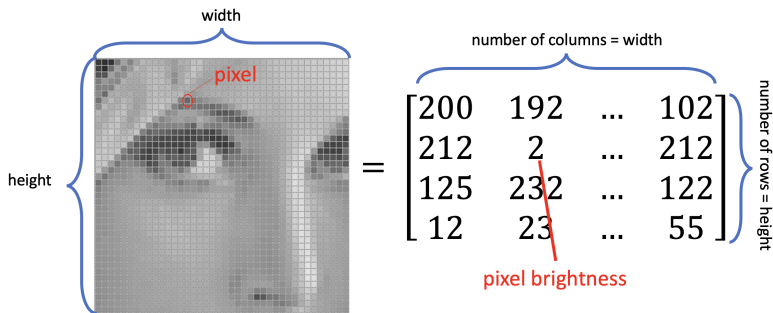


Tutorial 4: Displaying Images

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Images are Matrices



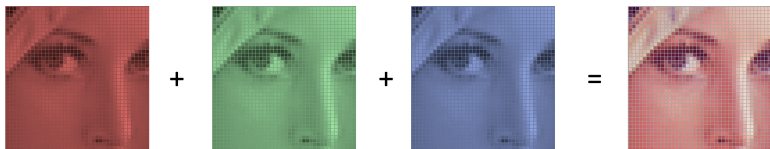
Grayscale images are expressed as matrices in computers.

A pixel in the image corresponds to an element in the matrix.

Each element of the matrix indicate the brightness of a pixel.

There are usually 256 levels of brightness, 0 is darkest while 255 is brightest.

Colored Images are Matrices too



One colored image is expressed as **three** individual matrices:

- Three matrices indicate brightness in Red, Green and Blue tones (RGB).
- Computer can display a colored image by stacking three images together.

Images Files are Flattened Matrices

Image files usually store images as flattened matrices.

- Many file system (such as tape) can only support sequential read/write.
- Recall, a row-major flattend matrix is

$$\begin{bmatrix} 1, & 2 \\ 3, & 4 \end{bmatrix} \Rightarrow [1, 2, 3, 4].$$

Knowing these facts, we can build a “textual image viewer” using C programming language.

Building an Image Viewer

Suppose you have obtained an `int` array `a` with length $M \times N$:

- It contains a flattened matrix $[12, 232, \dots, 254]$
- Let the “unflattened” matrix be $A \in \mathbb{N}^{m \times n}$.
- Matrix A represents an image with width N and height M .

To visualize your image, simply print out an M by N matrix replacing the integer value of $A_{i,j}$ according to the following rules:

- if $A_{i,j} \leq 85$, print empty space .
- if $85 < A_{i,j} \leq 170$, print character I.
- if $170 < A_{i,j} \leq 255$, print character M.

Building an Image Viewer

Complete `image2char` in `image2d.c`:

1. loop over every row and column;
2. access each pixel with `image[i * Width + j]`;
3. print the corresponding character;
4. print a newline after each row.

Then uncomment the call to `image2char` in `main`.

Keep `surprise.dat` in the same folder and run the program from that folder.

Can you see the image?

If not, please zoom out by pressing `ctrl + -`